

Solving Sudoku Using Integer Linear Programming

Abstract

This report presents a concise and complete integer linear programming (ILP) formulation for solving a standard 9×9 Sudoku puzzle. Sudoku's rules can be represented exactly using binary decision variables and linear equality constraints. This transforms the puzzle into a pure feasibility problem solvable by modern MILP solvers such as Gurobi.

1 Introduction

Sudoku consists of a 9×9 grid subdivided into nine 3×3 blocks. The task is to assign digits 1–9 to all cells so that:

- each row contains all digits exactly once,
- each column contains all digits exactly once,
- each 3×3 block contains all digits exactly once,
- given clues must be respected.

These constraints are linear and can be encoded using integer linear programming.

2 Decision Variables

Let

$$r, c \in \{1, \dots, 9\}, \quad d \in \{1, \dots, 9\}.$$

Define binary variables:

$$x_{rcd} = \begin{cases} 1, & \text{if cell } (r, c) \text{ contains digit } d, \\ 0, & \text{otherwise.} \end{cases}$$

There are $9 \times 9 \times 9 = 729$ binary variables.

3 ILP Constraints

3.1 Cell Constraints

Each cell must contain exactly one digit:

$$\sum_{d=1}^9 x_{rcd} = 1 \quad \forall r, c.$$

3.2 Row Constraints

Each digit must appear exactly once in each row:

$$\sum_{c=1}^9 x_{rcd} = 1 \quad \forall r, d.$$

3.3 Column Constraints

Each digit must appear exactly once in each column:

$$\sum_{r=1}^9 x_{rcd} = 1 \quad \forall c, d.$$

3.4 Block Constraints

For block indices $b_r, b_c \in \{0, 1, 2\}$, every digit must appear exactly once in each 3×3 block:

$$\sum_{r=3b_r+1}^{3b_r+3} \sum_{c=3b_c+1}^{3b_c+3} x_{rcd} = 1 \quad \forall d, b_r, b_c.$$

3.5 Given Clues

If the Sudoku puzzle provides a clue with digit d^* in cell (r, c) , then:

$$x_{rcd^*} = 1.$$

This automatically forces all other digits for that cell to zero via the cell constraint.

4 Objective Function

Sudoku solving is purely a feasibility problem, so no optimization is needed:

$$\max 0.$$

A MILP solver will find values of x_{rcd} satisfying all constraints, producing a valid Sudoku solution.

5 Reconstructing the Solution

Once solved, the digit assigned to each cell (r, c) is:

$$\text{Digit}(r, c) = d \quad \text{such that} \quad x_{rcd} = 1.$$

6 Conclusion

The integer linear programming formulation presented here provides a compact, exact, and mathematically elegant method for solving any standard Sudoku puzzle. With 729 binary variables and a few hundred linear equality constraints, modern MILP solvers can compute solutions quickly and reliably.